WILDLIFE GENETICS

THE TIGER CHANGING ITS STRIPES?

By Rohit Parihar



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It's seen such an unalterable fact of nature that it has yielded a proverb: a tiger cannot change its stripes. But that's exactly what India's iconic national animal is doing. Something curious is afoot in the slender pool of tigers left in our pockets of wilderness—and it's happening precisely because the pool is so slender. Inbreeding is inevitable, and its consequences are showing up in a startling visual metaphor. Gene mutations have led to tigers being born with wider black stripes in Odisha's Simlipal Reserve. These apparent freaks of nature—being called "black tigers"—have a precursor. White tigers, a dismal product of inbreeding, have been known in American zoos since the 1950s. Animal geneticists are worried. Such tigers typically have severe deformities such as cross eyes and cleft lips, besides high mortality rates. In Sariska, Rajasthan, a blind cub born two years ago—and dead within months—was ascribed to the genetic curse. The spectre of decay in isolated tiger populations due to mating between closely related individuals—called 'inbreeding depression'—is now driving what could be India's first attempt to intervene in the genetic diversity of the big cat.

Rajasthan is a hotspot. After local breeds were wiped out from Sariska Reserve circa 2004, tigers from Ranthambore have a monopoly over these wilds. So experts apprehend a risk of inbreeding depression in Rajasthan's entire tiger population. Three global research papers last year indicated as much. Solution? Planned translocations to improve genetic diversity. The idea is to shift a few tigers from Maharashtra's Chandrapur district, which officially has 200 of the state's 311 adult tigers, to Rajasthan. Wildlife officials from the two states have been in touch, and are awaiting a nod from the National Tiger Conservation Authority (NTCA), says Deep N. Pandey, head of Forest Force and principal chief conservator of forests and wildlife, Rajasthan. Then the tigers have to be monitored for two years before they are shifted.

As a source, Chandrapur suggests itself. Its 300+ population, including cubs and sub-adults, makes it the district with the highest number of tigers in India. The high density also poses a high risk of man-animal conflict, so local authorities are only too glad about the idea of exporting some. "We must move out some tigers from Chandrapur and Rajasthan is the best option as it would diversify the gene pool there," says Sunil Limaye, principal chief conservator of forests and chief wildlife warden, Maharashtra. Tigers can be a political symbol too, but with non-BJP governments in both states, Maharashtra CM Uddhav Thackeray is likely to give the go-ahead.

RANTHAMBORE TIGERS ARE FEARED TO BE SHOWING SIGNS OF 'INBREEDING DEPRESSION'

Caution will be the watchword, for the memory of failure haunts wildlife officials. India's first inter-state tiger relocation project that began in 2018—not for genetic rescue, but to augment population—had had a less than ideal outcome. A tiger shifted from Kanha in Madhya Pradesh to Odisha's Satkosia Reserve fell to poachers within months, while a tigress shifted from MP's Bandhavgarh was accused of killing two persons at Satkosia and was sent back in 2021 after being kept in a cage